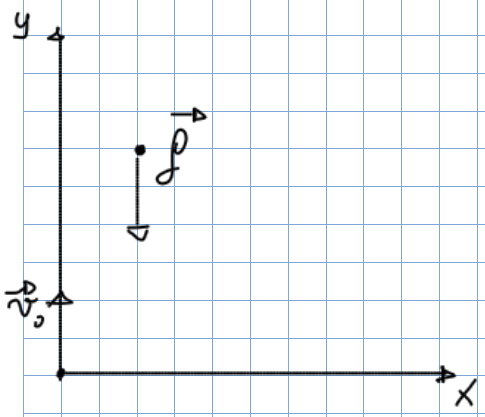


$$y(t) = y_0 + v_{0y}t + \frac{1}{2}at^2$$

$$y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$v_{0y} = v_0 > 0$$

$$y_0 = 0$$

$$\Big\} \quad y(t) = v_0 t - \frac{1}{2}gt^2$$

$$y'(t) = v_0 - gt = 0 \Rightarrow \frac{v_0}{g} = t_m$$

$v_0 = \frac{m}{s}$ $\quad g = \frac{m}{s^2}$

$$\frac{v_0}{g} = \frac{m}{s} \cdot \frac{s^2}{m} = \text{secondi}$$

$$y''(t) = -g < 0$$

$$y(t_m) = v_0 t_M - \frac{1}{2}g t_m^2 = \frac{v_0^2}{g} - \frac{1}{2}g\frac{v_0^2}{g^2} = \frac{v_0^2}{2g} = h_M$$

$$v_y(t_m) = 0$$

$$v_y = \left.\frac{dy}{dt}\right|_{t_m} = 0$$

↗ il tempo preciso in cui la pallina arriva a terra

$$v_y(t) = v_0 - gt$$

$$y(t_S) = 0 = v_0 t_S - \frac{1}{2}g t_S^2 = t_S\left(v_0 - \frac{1}{2}g t_S\right) \Rightarrow \begin{cases} t_S = 0 \\ t_S = \dfrac{2v_0}{g} = 2t_M \end{cases}$$

$$t_S = 2t_M = 2\frac{v_0}{g}$$

$$v_y(t_S) = v_0 - \frac{2v_0 g}{g} = v_0 - 2v_0 = -v_0 \Rightarrow v_y(t_S) = -v_0$$

↗ il modulo è uguale a quello di salita, il verso è opposto

$$|v_y(t_s)| = v_0$$

Cosa succede se partiamo da una quota h diversa da quella del suolo

cambia la condizione iniziale

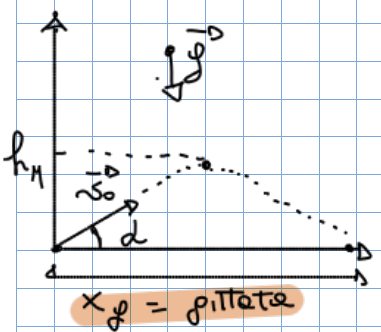
$$y(t) = h + v_0 t - \frac{1}{2} g t^2$$

$$v_y(t) = v_0 - g t$$

$$v_y(t_m) = 0 \Rightarrow v_0 - g t_M = 0 \Rightarrow t_M = \frac{v_0}{g}$$

$$y(t_m) = h + v_0 t_M - \frac{1}{2} g t_M^2 = h + \frac{v_0^2}{g} - \frac{1}{2} g \frac{v_0^2}{g^2} = h + \frac{v_0^2}{2g} = h_M$$

Moto parabolico



$$\vec{v_0} = (v_{0x}, v_{0y}) \qquad \vec{r_0} (P_{0x}, P_{0y})$$

$$\begin{cases} v_{0x} = v_0 \cos\alpha \\ v_{0y} = v_0 \sin\alpha \end{cases} \qquad P_{0x} = P_{0y} = 0$$

$$\begin{cases} x(t) = v_{0x} t = v_0 \cos\alpha\, t \\ y(t) = v_{0y} - \frac{1}{2} g t^2 \end{cases} \Rightarrow \begin{cases} v_x(t) = \frac{dx}{dt} = v_{0x} = v_0 \cos\alpha \\ v_y(t) = \frac{dy}{dt} = v_{0y} - g t = v_0 \operatorname{sen}\alpha - g t \end{cases}$$

$$v_y(t = t_M) = 0 = v_{0y} - g t_M \Rightarrow t_M = \frac{v_{0y}}{g} = \frac{v_0 \sin\alpha}{g}$$

$$h_M = y(t = t_M) = v_{0y} t_M - \frac{1}{2} g t_m^2 = \frac{v_0^2 \sin^2\alpha}{g} - \frac{1}{2} \frac{v_0^2 \operatorname{sen}^2\alpha}{g} = \frac{v_0^2 \sin^2\alpha}{2g}$$

$$t_g = 2 t_m = \frac{2 v_0 \sin\alpha}{g} \qquad 2 \sin\alpha \cos\alpha = \sin 2\alpha$$

$$x_G = x(t = t_G) = \frac{v_0 \cos\alpha \; 2 v_0 \sin\alpha}{g} = \frac{2 v_0^2 \sin\alpha \cos\alpha}{g} = \frac{v_0^2 \sin 2\alpha}{g}$$

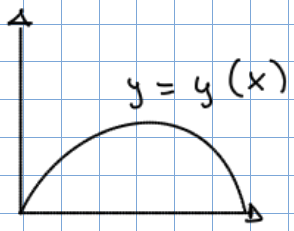
X_G MAX $\Leftrightarrow$ $\sin 2\alpha_{MAX} = +1 \Leftrightarrow 2\alpha_{MAX} = \frac{\pi}{2} \Leftrightarrow \alpha_{MAX} = \frac{\pi}{4}$

Le traiettorie

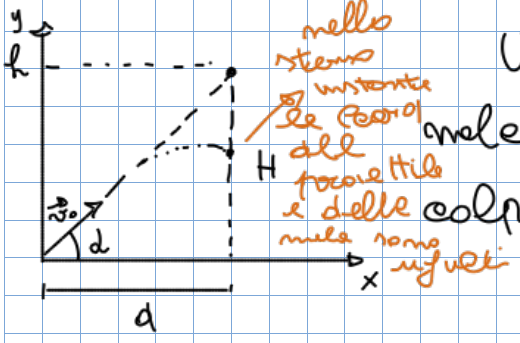
$x(t) = v_0 \cos\alpha\, t \Rightarrow t = \frac{x}{v_0 \cos\alpha} \Rightarrow y = \frac{v_0 \sin\alpha}{v_0 \cos\alpha} x - \frac{1}{2} g \frac{x^2}{v_0^2 \cos\alpha}$

$\tan\alpha$

$y = ax - bx^2 \quad \begin{cases} a > 0 \\ b > 0 \end{cases}$



Il problema del tiratore



Un uomo spara e allo stesso momento la mela si stacca dall'albero, riuscirà a colpirla lo stesso?

$\frac{H}{d} = \tan\alpha \Rightarrow H = (\tan\alpha) \cdot d$

$\begin{cases} y_M(t) = H - \frac{1}{2} g t^2 \\ x_M(t) = d \end{cases} \qquad \begin{cases} y_P(t) = v_0 \sin\alpha\, t - \frac{1}{2} g t^2 \\ x_P(t) = v_0 \cos\alpha\, t \end{cases}$

$t_d = \frac{d}{v_0 \cos\alpha}$

$y_P = (t = t_d) = v_0 \sin\alpha \frac{d}{v_0 \cos\alpha} - \frac{1}{2} g t_d^2 = (\tan\alpha) d - \frac{1}{2} g t_d^2$

$y_M(t = t_d) = H - \frac{1}{2} g t_d^2$

$H = (\tan\alpha) d$

uguale

$y_M(t = t_d) = H$

$$y_n(t=t_d) - y_p(t=t_n) = H - (t_{on}-1)d + \frac{1}{2}g t_d^2$$